

AIRBUS FOUNDATION



UNIVERSIDAD NACIONAL
AUTÓNOMA DE MÉXICO

**Morelos Soars: Charting New Horizons in Mexico STEM
education, Guided by the Airbus Foundation.**

Institution

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Co-Leader: Prof. Maribel Abarca, Director of COBAEM (Colegio de Bachilleres del Estado de Morelos). Prof. Abarca's leadership at COBAEM, which serves over 1,200 highschool students with declared interest in STEAM education (bachillerato in Mexico, Liceum in Europe), involves more 100 science teachers and is instrumental in reaching the target student population and ensuring the project's successful implementation within the state high school system. Her commitment underscores the strong institutional partnership for this initiative. There are other highschool systems in Morelos but ICF-UNAM and Cobaem have a longstanding and good collaborative track record.

1. Context

Morelos is a small state south of Mexico City, in Mexico. Despite its natural beauty, is a state of profound contrasts, facing significant socio-economic challenges, particularly in its marginalized rural and indigenous communities. This is not a distant problem; it is the daily reality for a staggering number of its citizens, specially the young. According to the latest official data from CONEVAL, the Mexican agency for human development monitoring, over 650,000 people in Morelos live in poverty, with more than 100,000 of them experiencing its most extreme forms. For young people in these regions, this is not just a statistic. It means that many families lack access to the most basic services, and students often face long, arduous journeys on foot just to reach their schools. It is a testament to their powerful will to learn that they make this journey every day to look for learning. These young people, aged 15 to 18, are acutely aware of the challenges surrounding them—water scarcity, agricultural difficulties, and limited opportunities. As many young people nowadays, they hold an clear understanding that science and technology are not abstract subjects, and motivated teachers of the COBAEM (one of several higschool institutions, which are part of the state system for public highschool education) system have reinforced this belief. In this context, we believe, powerful tools provided by a STEAM frame of mind can help them begin tackling their local problems. For these students, who attend distance learning high schools (EMSAD units affiliated with COBAEM) with limited resources, the opportunity to engage with the Airbus Foundation Discovery Space represents more than just access to information; it is a gateway to a new world of possibilities. It can be a truly transformative experience, providing them with the critical thinking skills, problem-solving methodologies, and innovative frameworks they need to build a better future for themselves and their communities. Our Institute of Physical Sciences, which is part of the National University of Mexico, possesses a robust capacity for high-quality scientific mentorship, yet our reach often does not extend to these underserved communities. This proposal addresses the urgent need to bridge this gap, not just by teaching science, but by fostering science with community impact, leveraging our scientific expertise in collaboration with young students and teachers of the public high school system in Morelos (COBAEM) to empower these bright, resilient young students and their motivated teachers. Although we already have had very enriching interactions (UNAM and COBAEM), and a friendly and professional liason with COBAEM, we have done so with limited resources and even personal contribution from the staff of our Institute. We truly believe the seed funding you may generously provide can make a big difference and will be a spark that may ignite new and more ambitious initiatives, one entusiastic step at a time.

2. Objectives

The overarching goal of this project is to foster interest and competence in STEM among young students ages 15 to 18 years old in Morelos, Mexico, and make them aware of the airbus foundation resources as a key source of inspiring knowledge and foster the spark of action through STEAM related projects. This will be achieved, we believe, through a unique synergy: combining the human-centered mentorship of our Institute, which as part of the National University (UNAM) , with the world-class digital resources of the Airbus Foundation Discovery (AFDS) Space platform. This will be carried out by making the Discovery Space platform accessible, initially, to a potential audience of 1,200 students with declared interest in STEAM areas, which, in the first year of the project will be focussed on 300-400 enthusiastic COBAEM students supported by approximately 40 teachers. The overall vision of this project is that we can empower them to start conceptualizing and proposing solutions real-world problems within their communities, motivated through the material of the (AFDS) and personal tutoring by scientists. The seed money will be invaluable to fund small initiatives at the level of concept or minimum viable experiments, as will be described later in this document. Since some of the schools are located at remote areas, we contemplate that some of the resources will also be devoted to allow them to reach access to the internet

through a satellite internet provider. We believe, at the ICF-UNAM that this project aligns directly with the Airbus Foundation's mission of youth development and fostering educational equity.

Our specific objectives are to:

- 2.1 **Empower young people with critical skills and knowledge (Airbus Foundation Impact Staircase Level 6):** We will provide participants with the tools and confidence to identify local challenges and develop tangible STEM-based solutions. The Airbus Discovery Space (AFDS) content is essential here, providing a structured framework and material developed specially to the age of the highschool students. We expect that they will use these modules as a "toolbox" of ideas and motivation, which our mentors will help them to mature and encourage them to propose potential solutions to pressing local issues. In this process, **COBAEM educators will work alongside our mentors**, gaining firsthand experience in project-based learning and learning how to guide students in adapting digital resources for local contexts, thereby enriching their own pedagogical practices.
- 2.2 **Enable young people to gain solid knowledge in STEM-related topics (Airbus Foundation Impact Staircase Level 5):** We aim to deliver a foundational and practical understanding of scientific principles. For students in remote areas, the Airbus Discovery Space serves as the high-quality scientific and friendly platform they sometimes lack. The access of (AFDS) combined with our outreach activities and workshops, will make the STEM related COBAEM teaching body at participating schools—**directly by the direct participation of 40 educators and with indirect exposure to approximately 100+ educators—aware of the Airbus Foundation Discovery Space**. Our goal is to actively engage **at least 40 of these educators as direct co-mentors who will be using and adapting the content** into their lesson plans, moving beyond the standard curriculum to foster deeper student understanding and passion for Stem subjects.
- 2.3 **Develop a strong curiosity and connection to STEM learning (Airbus Foundation Impact Staircase Level 4):** We will ignite a lasting passion for scientific inquiry through a hands-on hackathon guided by voluntary collaborating scientists from the Institute of Physical Sciences that will not charge money to the project. We believe The (AFDS) are the catalyst, showcasing the "wow-factor" of modern science. This content will be introduced by their teachers, who will be trained to use it as a springboard for imagination in our state-wide competition, inspiring students to ask "what if?" The most exciting ideas will then be developed during the hackathon, transforming the initial spark of curiosity ignited by the Discovery Space (AFDS) into a flame of hands-on creation.
- 2.4 **Cultivate young people as global citizens (Airbus Foundation Impact Staircase Level 7):** Through project-based learning, we will enhance students' problem-solving abilities and help them connect their local actions to global challenges. The Airbus Foundation's content inherently provides this global perspective. When students use modules on satellite imaging to monitor local deforestation, they are participating in a global conversation. This provides a replicable model for educators to frame their own STEM teaching within a context of global citizenship and local impact.
- 2.5 **Strengthen the Educational Ecosystem through Educator Empowerment and Institutional Partnership:** Beyond individual student and teacher impact, this project is designed to foster systemic change. We will establish a "Community of Practice" for the participating educators, enabling them to continuously innovate and share best practices for integrating the Airbus content. This professional development will equip them to respond to emerging student needs long after the project funding ends, hence the impact of the (AFDS) material will be repeated over the next year students these teachers had. Furthermore, this initiative solidifies a strategic partnership between our research institution and the state's public high school system, creating a permanent channel for knowledge exchange and ensuring that the project's impact leads to lasting institutional improvements in STEM education for the entire region. We plan, may we receive the seed funding, to give ample publicity

to the generous patronage of the Airbus Foundation, make a wide diffusion of the (AFDS) material and seek for more resources beyond the seed money you may generously provide.

3. Target Audience, direct and indirect

The purpose of this section is to offer a clear picture of the scope and both direct and indirect impacts of the (AFDS) material and the of beneficiaries of the program and to focus on the single year of operation that the seed funding provides. The total student population devoted to STEAM related studies of the COBAEM system in Morelos is more than 1,200 students, of a much larger universe of students that focus more on social sciences, or technical interest on other areas (linguistics, agriculture, etc). This is according to personal communication with the COBAEM partners a universe of (708) women and (497) male students. This project is designed to the most active part of this population as direct beneficiaries. According to our COBAEM partners, this active student that responds to calls for activities and events and which is, usually between 300 and 400 per year and is supported by 40 teachers. We estimate that the indirect beneficiaries in the first year, through the dissemination of the project, public presentation of the result of the workshops and projects, along with media publication of the project and but will be at least the 1,200 population of those that have declared being inclined towards STEAM fields. Given the prestige of Airbus and the local appreciation of the National University of Mexico, these numbers are estimated at a very conservative level. Let us not forget that, after the year of the seed funding, the teachers that participated will receive a new generation of students; hence the impact will go beyond the first year, but it is not easy to estimate these longer term indirect beneficiaries.

To provide more details we define our reach, on the first year, as follows:

- **Direct Beneficiaries (Intensive Participation):** This is the core group that will receive the most in-depth and comprehensive experience.
 - **300-400 students:** These are the members of the 30-40 teams that will be selected to participate in the hackathon. These students will be required to consult the (AFDS) material and will receive months of intensive, personalized mentorship from the scientists at our institute.
 - **40 COBAEM educators:** This group of teachers will participate as direct co-mentors, receiving hands-on training in project-based learning and in integrating AFDS content into their lesson plans.
- **Secondary Beneficiaries (Initial Exposure):** This group, comprising the whole 1,200 STEM focussed students at COBAEM, will interact significantly with the 300-400 active participants of the program, by attending the presentations and pitches of the selected students and will thus be made aware of the AFD amterial, though not with the same intensity as the core group.
 - **Second Beneficiaries (Teachers) Up to 150 educators:** We estimate this level of participation of other teachers in STEAM releted areas in the presentations of the projects, presentation of the selected students and introductory virtual workshops that will launch the statewide competition, this in addition to the direct 40 teachers related directly with the mentoring of the selected students.
- **Indirect Beneficiaries (Ecosystem Impact):**
 - **The COBAEM student body and other public and private funded schools:** The whole student community of highschool students in Morelos is in the tens of thousands. We believe this population will benefit, albeit in the long term, from the dissemination of the initial

competition, the projects and final "STEM Impact Fair," and, in the long term, from the new pedagogical methods that trained educators will made aware of through this initial program. We believe this type of initiatives, although initially modest in reach on the first year may help creating systemic change in STEM education throughout the region. This objective goes beyond the 1 Year duration of the initial project that will be funded through the seed money, we passionately believe in this, at the ICF-UNAM.

4. Specific usage of Airbus Foundation Discovery Space materials

We are fully aware that a fundamental part of this seed funding is to promote the active visualization and comprehensive usage of the Airbus Foundation Discovery Space (AFDS) material. Our project has been designed from the ground up to guarantee that the AFDS platform is not merely a supplementary resource, but a central part of the pedagogical tools driving every activity. This section explains our specific strategy for Content Utilisation, detailing how the selection and use of AFDS modules, videos, and experiments will be methodically woven into the fabric of our workshops and competition phases. We hereby guarantee the content's foundational place, ensuring its educational impact is maximized and that it serves as the primary driver of learning, innovation, and inspiration for our students. In what follows we will give some examples of how would we plan to use the AFDS material. The final way we do this will be guided by other ideas that the students and teachers may pose, but this is an indication on the general strategy and ideas.

- **Problem Identification through Earth Observation:** The first workshop will require students to use the downloadable activity sheet from the "**Satellites are fascinating**" module, "Become a satellite image analyst." Using publicly available satellite (E.G. Landsat) data of their own communities, they will identify a tangible local problem—such as soil erosion, water reservoir depletion, or an unauthorized landfill—providing a data-driven foundation for their proposal.
- **Connecting Local Issues to Global Goals:** We will screen the "Sustainability" video from the "**Play with Science**" section, which introduces Sustainable Development Goals (SDGs). Students will then be tasked with mapping their satellite-identified problem to a specific SDG, framing their local challenge within a global context from day one.
- **Inspiration for Tangible Solutions:** To spark creative solutions, we will showcase specific videos as case studies. For energy projects, we will analyze the "ZEROe" concept aircraft from "**Future of the skies**," focusing on hydrogen power, and challenge students to think how to design a small-scale hydrogen production system for their school lab, using the "How to produce hydrogen?" experiment as a guide. In our institute we have catalysis experts that could help guiding this in practical terms. For agricultural projects, we will draw inspiration from the sustainable lifecycle of biofuels (SAF) to brainstorm ways to convert local agricultural waste into energy for water pumps, albeit in the conceptual stage.
- **Hackathon Workgroups:** During the intensive hackathon phase, the AFDS platform will function as the official technical manual and virtual laboratory for the selected teams.
 - **Structured Experimentation as the Core Activity:** The hackathon will not be an unstructured build; it will be a series of guided workshops based on the "**Stimulate your scientific mind**" experiments. Teams focused on aerodynamics (e.g., designing a drone for crop dusting) will begin by building and testing principles from the "How does air flow around a wing?" video. Teams working on environmental monitoring will first build the DIY compass to understand orientation for their sensors. These downloadable activity sheets will serve as their official laboratory manuals.

- **Cross-Disciplinary Innovation:** We will explicitly encourage students to apply concepts from one domain to solve a problem in another. For instance, a team tasked with creating a small, all-terrain robot to monitor crop health will be directed to the **”Mission to the Moon”** module. They will analyze the lunar rover’s articulated suspension system and autonomous navigation logic to design a robust agricultural robot capable of navigating uneven local terrain. This directly transfers principles of space exploration to solve a terrestrial, community-level problem.
- **Direct Mentorship and Impact Communication:** Mentorship will extend beyond technical guidance to cultivate critical thinking and compelling communication, using the AFDS **”Inspire”** series as a structured pedagogical tool.
 - **Targeted Problem-Solving:** When a team encounters a technical hurdle, mentors will not provide the answer but will guide them to a specific piece of AFDS content. For example, if a team’s sensor-deployment prototype is failing, the mentor will say, “Let’s watch the video on helicopter rescue hoists in **’Helicopters in emergency situations.’** Analyze the winch and cable-guiding mechanism. How can we adapt that design to ensure your sensor deploys without tangling?”
 - **Structuring the Narrative for Impact:** To prepare for the final **”STEM Impact Fair,”** we will conduct a mandatory communication workshop modeled on the animated series **”FAIR ENOUGH.”** Students will engage in a structured debate, role-playing as the characters Lila and Martin, to argue the ethical and social implications of their own projects. This forces them to think beyond the technology and articulate its human impact.
 - **Fostering Personal Growth and Vision:** In the final stages, each student will work through the exercises in **”The guide to becoming yourself.”** Their final presentation will require them to answer a question from the guide: **”What skills did you discover in yourself during this project?”** This integrates their personal journey into the technical presentation, teaching them to articulate not just what they built, but who they became in the process.

Direct Mentorship and Impact Communication: Mentorship will focus on both technical execution and developing the student’s voice, fully integrating the AFDS **”Inspire”** and **”Experiment”** content.

- **Problem-Solving with Mentors:** Instead of giving answers, a mentor will guide a struggling team by saying, “Let’s review the video on landing airplanes in the **’Science of Flight’** section. The principles of drag and braking could help us design a better parachute for your water-quality sensor drop.” This method empowers students to find solutions using the AFDS resources, reinforcing its value.
- **Developing a Powerful Narrative:** To prepare for the final **”STEM Impact Fair,”** we will move beyond technical skills and use the **”FAIR ENOUGH”** series to teach critical thinking and communication. In addition to this, mentors will lead workshops based on the series’ methodology, prompting teams to debate the social and ethical implications of their solutions. Furthermore, we will use the **”The guide to becoming yourself”** booklet to help students connect their project to personal goals and future aspirations, allowing them to present not just a prototype, but a compelling vision for their community’s future and their role in it.

5. Specific Activities

The project will unfold in four distinct phases, integrating the Airbus Foundation Discovery Space digital content as its core.

Phase 1: Discovery, Ideation, and Competition (Months 1-3)

- 5.1.1 **Guided Exploration of Airbus Foundation Discovery Space:** Students will be guided through selected modules of the Discovery Space. An online competition will challenge them to use this knowledge to draft a one-page proposal for a STEM project solving a local community problem (e.g., water quality, agricultural efficiency, sustainable energy).
- 5.1.2 **Selection of Hackathon Teams:** A joint committee of Institute scientists and COBAEM teachers will select the top 30-40 proposals. The selected students will form teams to participate in the main project phase.

Phase 2: Mentored Hackathon and Project Development (Months 4-8)

- 5.2.1 **Intensive Mentorship:** This phase involves intensive mentorship by scientists from our Institute. Mentors will visit the communities to guide students through practical, hands-on project development.
- 5.2.2 **Community-Centered Hackathon:** A central 3-day "Hackathon for Social Impact, based on STEAM solutions" event will be held. Teams will work intensively to build prototypes, run simulations, or finalize data analysis for their projects. The Discovery Space content will serve as a foundational technical resource throughout.
- 5.2.3 **Suggested Project Areas:** Observational Astronomy, Instrumentation with Raspberry Pi/Arduino for environmental monitoring, and Modeling/Simulation for local physical phenomena among others suggested by local highschool teachers and students.

Phase 3: Community Showcase and Dissemination (Month 9)

- 5.3.1 **Local STEM Impact Fair:** A culminating "Impact Fair" will be organized in a central community location. Students will present their finished projects to peers, families, teachers, and local authorities, demonstrating the real-world impact of their work, commenting how the Airbus foundation knowledge base and seed money made a difference between dreaming and actually starting to propose solutions.

Phase 4: Evaluation and Reporting (Months 10-12)

- 5.4.1 **Final Evaluation:** Comprehensive project evaluation through post-project questionnaires, analysis of project portfolios, and interviews.
- 5.4.2 **Final Report:** Preparation and submission of the final report to the Airbus Foundation.

6. Expected Outcomes and Impact

Based on the Airbus Foundation Impact Staircase and our dual-empowerment model for students and educators, we anticipate the following specific outcomes:

- 6.1 **Level 7: Young people become global citizens equipped with critical skills.** *Outcome:* Participants will not only enhance their critical thinking but will also learn to contextualize local problems within a global framework. They will move from identifying a community issue to designing a viable, science-based solution for it. *Metric:* At least 80% of participating student teams will successfully develop a functional prototype or a detailed simulation addressing their chosen local challenge. Crucially, in their final presentation, they must be able to articulate the project's broader implications by referencing at least two relevant global concepts (e.g., climate resilience, sustainable development) drawn from the Airbus Discovery Space modules they consulted.
- 6.2 **Level 6: Young people are empowered with the knowledge to take action.** *Outcome:* Students will gain genuine confidence and a sense of agency, understanding that they possess the skills to apply scientific knowledge to real-world scenarios. This shifts their perspective from being passive recipients of education to active agents of change in their communities. *Metric:* Post-project surveys will show a 70% increase in students' self-reported confidence in their ability to use STEM to solve problems. This will be supported by qualitative data from interviews, where students will be asked to provide specific examples of how their approach to problem-solving has changed.
- 6.3 **Level 5: Young people gain solid knowledge in STEAM-related topics.** *Outcome:* Participants will acquire a deep, practical understanding of core scientific principles directly relevant to their projects, learned through active application rather than passive memorization. This new knowledge will be co-facilitated by their teachers, who will also be learning to apply these concepts. *Metric:* Performance on post-project technical assessments, tailored to the specific Airbus Discovery Space modules used by the teams, will demonstrate an average 60% improvement in understanding k

7. Evaluation on how well we did in achieving AFDS materials

We plan a comprehensive, mixed-methods evaluation strategy designed to measure the project's effectiveness against each of our stated objectives. This approach will provide both quantitative data and the rich qualitative stories needed to assess the true impact on students, educators, and the wider community.

7.1 Pre/Post-Project Questionnaires for Students: To quantitatively measure shifts in attitudes and knowledge, we will administer carefully designed surveys at the beginning and end of the program. These will assess:

- a) **Confidence and Agency (Impact Level 6):** Using Likert-scale questions to measure self-reported confidence in using STEM to solve community problems, aiming for our target of a 70% increase.
- b) **Conceptual Knowledge (Impact Level 5):** A short, baseline technical quiz focused on core principles from the most relevant Airbus Discovery Space modules. A post-project quiz will measure progress toward our goal of a 60% average improvement.
- c) **Career Aspirations (Impact Level 4):** Questions tracking changes in student interest towards pursuing higher education or careers in STEM fields.

7.2 Educator and Mentor Feedback and Reporting: We will evaluate the project's impact on educators through multiple channels:

- a) **Structured Debriefing Sessions:** At key milestones (post-competition, mid-hackathon, post-fair), we will hold guided feedback sessions with the cohort of 40+ participating COBAEM teachers to discuss challenges, successes, and the practicalities of integrating the Airbus content.
- b) **Teacher Practice Surveys:** A specific survey for educators will measure their confidence in using project-based learning and digital resources like the Discovery Space, providing data on the effectiveness of our "Community of Practice" and training efforts.

7.3 Direct Assessment of Airbus Foundation Content Integration: To ensure the core requirement of the grant is met and measured, we will evaluate the usage of the Discovery Space through:

- a) **Project Portfolios and Rubrics:** The final evaluation of student projects will use a detailed rubric where "Effective Use of Airbus Discovery Space" is a key criterion. Teams will be required to document which modules they used as a guide and how it informed their design choices. This directly links their final product to the platform's resources.
- b) **Usage Analytics:** Where feasible, we will track engagement metrics and student pathways through the Discovery Space platform to understand which content is most impactful and where students require more guidance.

7.4 Qualitative Interviews and Storytelling: To capture the human impact behind the data, we will conduct semi-structured interviews with a selected sample of:

- a) **Students:** To hear their personal stories of discovery, challenge, and empowerment.
- b) **Teachers:** To understand how this project has transformed their pedagogical approach and their view of their students' potential.
- c) **Community Leaders:** To gather testimonials on the perceived value and inspiration generated by the final "STEM Impact Fair."

- 7.5 **Community Impact Assessment:** The final student projects themselves are a primary evaluation tool. The portfolios will be assessed by a joint committee of our scientists and COBAEM educators based on four core criteria: **Scientific Accuracy, Community Relevance and Impact, Innovative Use of Airbus Content,** and **Clarity of Communication,** directly measuring the project's success in achieving our highest-level outcomes.

8. Planning

The project is planned for a 12-month duration, structured into four distinct phases. This timeline ensures a logical progression from broad engagement to deep, mentored project development, culminating in a community showcase and a robust evaluation process.

Phase 1: Discovery, Ideation, and Engagement (Months 1-3)

This initial phase is focused on launching the project, generating widespread enthusiasm among the 1200 -strong COBAEM student body, that will ultimately help us selecting the 300-400 students that will directly participate by identifying the most motivated participants.

- 8.1.1 **Month 1: Kick-off and Strategic Outreach.** We will formalize our partnership with COBAEM, establish a joint steering committee, and finalize the rules for the "Solve a Local Problem" competition. This includes developing promotional materials and administering the pre-project baseline questionnaires to students and educators.
- 8.1.2 **Month 2: Competition Launch and Digital Immersion.** We will launch the state-wide competition, supported by introductory virtual workshops for all interested students and the 40+ science teachers. These sessions will provide a guided tour of the Airbus Discovery Space, demonstrating its potential and showing educators how to use it as a teaching tool.
- 8.1.3 **Month 3: Selection and Team Formation.** The steering committee will evaluate all submitted proposals based on feasibility, community impact, and effective use of Airbus content concepts. We will select the top 30-40 proposals, form the student teams, and assign a dedicated mentor from our Institute to each team, officially kicking off the mentorship journey.

Phase 2: Intensive Mentorship and Project Development (Months 4-8)

This is the core phase of the project, where teams transform their ideas into tangible prototypes with dedicated guidance from their mentors and the Airbus Discovery Space as their technical library.

- 8.2.1 **Months 4-5: Foundational Work.** Teams will have their initial meetings with mentors to refine their one-page proposals into detailed project plans. Mentors will guide students to the specific Discovery Space modules essential for their project (e.g., 'Coding' for an Arduino sensor, 'Earth Observation' for an environmental project). Project materials will be distributed, and teams will begin initial research and prototyping.
- 8.2.2 **Months 6-7: Deep Development and the Central Hackathon.** Mentorship will intensify with bi-weekly sessions. The centerpiece of this period will be the 3-day "Hackathon for Social Impact." This immersive event will bring all teams together to work intensively on their projects,

overcoming major technical hurdles with on-site support from our scientists and their COBAEM teachers.

- 8.2.3 **Month 8: Refinement and Data Analysis.** Following the hackathon, teams will focus on refining their prototypes, collecting and analyzing data, and interpreting their results. They will begin to structure their findings and prepare the narrative for their final presentations, learning how to communicate their scientific process effectively.

Phase 3: Community Showcase and Dissemination (Month 9)

This phase is dedicated to celebrating the students' achievements and sharing their innovative solutions with the communities they are designed to serve. Small but meaningful prizes may be offered here to motivate the most committed teams.

- 8.3.1 **Weeks 1-2: Final Preparations.** Students will put the final touches on their projects, posters, and presentations. Mentors will conduct "dress rehearsal" sessions, providing critical feedback on scientific communication and presentation skills to ensure every team is confident and prepared.
- 8.3.2 **Weeks 3-4: The STEM Impact Fair.** We will host a culminating "STEM Impact Fair" in a central community location, inviting families, local leaders, COBAEM officials, and the public. An expert panel will evaluate the projects, and an awards ceremony will celebrate the students' hard work, fostering a powerful sense of pride and accomplishment.

Phase 4: Evaluation, Reporting, and Future Planning (Months 10-12)

The final phase focuses on rigorously evaluating the project's impact, fulfilling our reporting duties, and laying the groundwork for long-term sustainability.

- 8.4.1 **Month 10: Comprehensive Data Collection.** We will administer the post-project questionnaires to all participants, conduct the semi-structured interviews, and collect all project portfolios for final assessment against our evaluation rubric.
- 8.4.2 **Month 11: Analysis and Dissemination.** Our team will analyze all collected data to synthesize key findings and learnings. We will draft the comprehensive final report for the Airbus Foundation and begin developing the open-source "Community STEM Hackathon Toolkit" to share our successful model.
- 8.4.3 **Month 12: Final Reporting and Strategic Continuation.** We will submit the final report to the Airbus Foundation. Concurrently, we will hold a strategic planning summit with COBAEM leadership, using the project's powerful results and success stories to begin the advocacy process for securing long-term state-level funding.

9. Budget

We request a total budget of €10,000 to cover the following categories for a focused pilot program:

- 9.1 **Project Materials (€4,000):** Raspberry Pi kits, Arduino boards, sensors, and basic tools for a smaller number of focused student teams. Justification: Essential for hands-on project construction.
- 9.2 **Logistical Support for Mentors (€2,000):** Travel expenses for scientists for a few key in-person workshops and the final event, supplemented by virtual mentorship. Justification: Facilitates crucial direct engagement in a cost-effective manner.
- 9.3 **Hackathon & Impact Fair Costs (€1,500):** Costs for a smaller-scale, locally-hosted event, covering basic materials, snacks, and small awards. Justification: Ensures the core collaborative and dissemination events can proceed.
- 9.4 **Connectivity Solutions (€1,000):** Provision of mobile hotspots or data packages for the most disconnected communities. Justification: Directly addresses a critical barrier for the target audience.
- 9.5 **Program Promotion and Outreach (€500):** Digital outreach and printing of materials for local schools. Justification: Ensures broad participation through efficient means.
- 9.6 **Contingency (€1,000):** 10% of the total budget for unforeseen expenses.

Total Requested Budget: €10,000

10. What we plan to do beyond this? Replicability, and a Vision for National Impact

This project is designed not as a one-time intervention, but as a scalable and replicable blueprint for educational transformation that can extend far beyond the borders of Morelos. Sustainability is our core consideration, and our plans are built on a strategy for long-term, systemic change.

The challenges faced by students in Morelos are, unfortunately, not unique. They mirror the reality for a significant portion of the millions of young people aged 15-18 living in other Mexican states. In places like Chiapas, Guerrero, and Oaxaca, where youth poverty rates are among the highest in the nation, countless bright students face similar barriers: limited resources, geographic isolation, and a lack of opportunities to apply their intelligence to the problems they see every day. This project is a direct answer to that widespread need. Since you ask us for ideas on how we could try to expand this project, let us dream a bit. We are well aware that dreaming has to be bound by realistic evaluations, but the seed planted by your seed funding may yield a generous tree, as the one in Oaxaca, the Tule, a beautiful 2000 year old tree in Oaxaca Mexico.

- 10.1 **Phase 2: Securing State-Level Support.** Armed with the project's proven impact, we would try to formally approach the Morelos State Government and local funding agencies. With a concrete evidence provided by your seed project, we could try to engage their partnership in transforming this initiative from a pilot into a recurring, state-funded annual event, making it a cornerstone of Morelos' educational strategy. An established Alumni Network will be crucial here, providing testimonials and peer mentors for future cohorts and the already trained teachers would be invaluable partners that could make this dream possible.
- 10.2 **Phase 3: National Expansion.** Why stop dreaming? Assuming we could engage state authorities, let us not forget that our University is a National one, albeit with several economic hardships. Why not thinking, if this project shows successful recurring events that we may try to leverage the validated model to showcase this project to other states and to national bodies?. The early endorsement of the Airbus Foundation, and our hard work with teachers and students would be

our "door-opener," lending credibility and demonstrating that this model meets the highest international standards. By presenting our toolkit and results, we may provide a clear roadmap for states like Oaxaca and Chiapas to adopt this program, exponentially widening the audience for the Airbus Foundation Discovery Space and bringing its invaluable content to tens of thousands more students across Mexico. Our ultimate aim is to secure federal and large-scale foundation funding to support this expansion over the next five years, explore funding agencies in the USA and Europe and give a longer scope to this.

We have laid out an ambitious vision, but we are fully aware that every great journey begins with a single, decisive step. For us, that first step is this proposal. We know the challenges are real, but the potential of these young students is far greater. We sincerely hope you evaluate this proposal positively and grant us the seed you are offering. We don't just see it as funding; we see it as a seed of trust. With the Airbus Foundation as our partner, we are ready to plant it, nurture it, and grow it into a strong tree—a tree with roots deep in our communities, and whose fruit will be the world-changing ideas of a new generation of leaders. The plans are ready, the passion is here, and the talent in Morelos is waiting to be unleashed. Please, allow us the opportunity to dream and to work, together with you, to make this inspiring vision a reality. Muchas gracias!.